

Machining

Instructions

VOLTA CONFLUENCE ENGINE – FABRICATION PARAMETERS PER METHOD

PARENT DOCUMENTS	LICENCE	CONCEPT
VCE-MFR-001 · VCE-CPR-001	CC BY-NC	Jake Galm, researched & documented 2026

ALL INSTRUCTIONS IN THIS DOCUMENT ARE ROUGH DRAFT

Every machining parameter in this document — feed rates, spindle speeds, depths of cut, toolpath strategies, cutter specifications — is **ROUGH DRAFT** unless explicitly labelled otherwise. Geometries and dimensions are engineering estimates based on design analysis. Fabrication parameters are indicative starting points derived from standard workshop practice for these materials. They must be reviewed and verified by a qualified operator with experience on the specific machine being used before any production parts are cut. A builder who treats these parameters as confirmed without that verification accepts full responsibility for the outcome.

To upgrade any instruction from Rough Draft to Confirmed: have a qualified machinist or fabricator review the specific parameters for their specific machine and material batch, run a test piece, and sign off on the parameters. Document the sign-off.

ELECTRONICS – QUALIFIED TECHNICIAN REQUIRED

All electronics assembly instructions in Section V of this document involve

circuits operating at up to 600V DC. Assembly, wiring, and commissioning of all high-voltage circuits must be performed by or under the direct supervision of a qualified electronics technician. Do not energise any high-voltage circuit without a bleed resistor (1M Ω , 5W) permanently connected across the capacitor bank output. Test all circuits with a dummy load before connecting valves, speakers, or any other load.

This document is organised by fabrication method. All parts fabricated by the same method are grouped together so a builder can batch their workshop time efficiently. Cross-reference each instruction to the Custom Parts Register (VCE-CPR-001) for full dimensional specifications and tolerances.

SECTION I

CNC *Milling*

One part requires CNC milling as its primary fabrication method. CNC turning (lathe work) is covered in Section II.

VCE-T01 — Turntable Platter × 2

ROUGH DRAFT

CNC Milling Instructions — 6061-T6 Aluminium

The platter is the most geometrically complex self-fabricated part. It requires accurate concentricity between the spindle bore, the bearing ring recess on the underside, and the outer diameter. Any eccentricity between these features causes vibration correlated to platter rotation — audible as wow through the signal chain.

Material Preparation

- 1 Source 6061-T6 aluminium round bar or plate, 310mm minimum diameter, 40mm minimum thickness. Verify alloy certification — do not use unknown-alloy scrap for this part.
- 2 Inspect for porosity or inclusions before machining. Cast aluminium is not acceptable — wrought 6061-T6 only.
- 3 Allow material to reach workshop temperature before starting — thermal expansion during machining will affect bore tolerance.

Machining Sequence

- 1 **Face both surfaces.** 0.5mm finishing pass. Establish perpendicularity reference. Target flatness: 0.05mm across full face.
- 2 **Rough outer profile.** Mill to 297mm diameter, leaving 1mm finishing allowance. 3mm DOC, 8mm 4-flute end mill, flood coolant. ROUGH DRAFT
- 3 **Bore spindle hole.** Drill pilot, bore to 9.8mm, ream to 10mm H7. Reaming must be done in a single pass without repositioning — concentricity of bore to outer diameter is critical. Tolerance: $\pm 0.01\text{mm}$ on diameter, roundness $< 0.005\text{mm}$.
- 4 **Flip and fixture on spindle bore.** All underside features must be machined from this datum to ensure concentricity. Arbor through 10mm bore, clamp on underside face.
- 5 **Machine bearing ring recess.** Dimensions from bearing ring OD — measured after procurement. Depth: bearing ring sits flush with platter underside $\pm 0.1\text{mm}$. Concentric with spindle bore to within 0.02mm.
- 6 **Machine eddy current disc mount.** Flat boss on platter underside, centred on spindle bore, diameter and thread to match disc fixing method. ROUGH DRAFT –
CONFIRM FIXING
METHOD BEFORE
MACHINING
- 7 **Finish outer diameter.** 0.2mm finishing pass to 295mm. 6mm end mill, dry. Aim for $1.6\mu\text{m}$ Ra on outer diameter.
- 8 **Machine felt mat recess on upper face.** 285mm diameter, 1.5mm deep. Blend to outer rim with 2mm radius. This is aesthetic — tolerance $\pm 0.3\text{mm}$ acceptable.
- 9 **Deburr all edges.** 0.3mm chamfer on all exposed edges. Remove all swarf from recess features.
- 10 **Dynamic balance.** Mount on arbor, run on dynamic balancing machine. Target: $< 1\text{g}\cdot\text{cm}$. Remove material from the outer rim in small increments — 2mm diameter holes drilled radially, maximum 5mm deep, evenly distributed around circumference. Do not remove material from the bearing ring recess area.

Machining Parameters ROUGH DRAFT – VERIFY WITH OPERATOR

OPERATION	PARAMETER	NOTES
ROUGHING – ALUMINIUM	2,500 RPM · 800mm/min · 3mm	Adjust for specific machine. Watch for chatter on large

	DOC · flood coolant	<i>DOC passes.</i>
FINISHING – ALUMINIUM	4,000 RPM · 400mm/min · 0.2mm DOC · dry	<i>Dry finish gives better surface on aluminium than coolant.</i>
REAMING – 10MM H7	400 RPM · 50mm/min · flood coolant	<i>Slow speed, light feed for accurate bore. Single pass — do not retract and re-enter.</i>
BEARING RECESS	3,000 RPM · 300mm/min · 0.3mm finishing DOC	<i>Interpolate with end mill rather than face mill for concentricity.</i>

FIRST ARTICLE CHECKS – BEFORE MAKING THE SECOND PLATTER

- ✓ Spindle bore diameter: 10.000mm–10.018mm (H7)
- ✓ Bore roundness: <0.005mm
- ✓ Bearing recess concentricity to bore: <0.02mm TIR
- ✓ Dynamic balance: <1g·cm on balancing machine
- ✓ Platter spins freely on ceramic shaft with no audible contact or roughness
- ✓ Float height with magnetic bearing installed: 1.2–1.8mm

DO NOT PROCEED TO SECOND PLATTER UNTIL FIRST ARTICLE PASSES ALL CHECKS

The second platter must be made to the same verified parameters as the first. If first article reveals any issues with the machining sequence, resolve them before starting the second part. Both platters must be interchangeable.

SECTION II

Lathe *Work*

The majority of self-fabricated parts are turned on a lathe. A lathe with 300mm swing and 600mm between centres is sufficient for all parts in this section. A 4-jaw independent chuck is required for the eccentric features on the gravity weight (VCE-OC01). All parts require deburring and edge chamfering after machining.

Lathe Instructions — OFHC Copper C10100

OFHC copper machines differently from aluminium and steel. It is soft, gummy, and prone to built-up edge on the tool face. Sharp HSS tools outperform carbide for finish work on OFHC copper. Flood coolant causes work-hardening and chatter in some operations — test on a scrap piece first.

Machining Sequence

- 1 Source OFHC copper round bar, 85mm minimum diameter, 10mm minimum length. Verify C10100 or equivalent — do not substitute ETP copper (C11000), which has lower conductivity.
- 2 **Face both sides.** Sharp HSS tool, no coolant. 0.3mm finishing pass. Target flatness: 0.05mm. Target surface finish: 0.8µm Ra — smooth faces maximise eddy current generation.
- 3 **Turn outer diameter to 80mm.** 3-jaw chuck on bar stock, face off, turn to diameter. Leave 0.3mm for finishing pass.
- 4 **Finish outer diameter.** 80mm ±0.1mm. Polish with 600-grit abrasive cloth on lathe — copper polishes well and the smooth rim reduces windage.
- 5 **Drill and bore spindle hole.** Centre drill, pilot drill, bore to 9.8mm, ream to 10mm H7. Copper is soft — use sharp reamer with cutting oil, single slow pass.
- 6 **Static balance check.** Mount on arbor, check static balance. Copper is dense — small asymmetries have measurable effect. Remove material from rim if needed: small chamfers on the heavier side.

Machining Parameters ROUGH DRAFT

OPERATION	PARAMETER	NOTES
ROUGHING – OFHC COPPER	600 RPM · 0.2mm/rev · HSS tool · no coolant	<i>Slow speed prevents built-up edge. Positive rake HSS tool essential.</i>
FINISHING – FACES	800 RPM · 0.05mm/rev · 0.3mm DOC · HSS	<i>No coolant on finishing pass. Flood coolant can cause chatter on copper.</i>

REAMING – 10MM H7

200 RPM ·
30mm/min ·
cutting oil

*Slower than aluminium —
copper grabs reamers if fed
too fast.*

ACCEPTANCE CHECKS

- ✓ Both faces parallel to within 0.03mm
- ✓ Face surface finish: 0.8µm Ra or better (smooth to fingertip)
- ✓ Bore: 10.000–10.018mm H7
- ✓ Disc fits spindle snugly with no radial play
- ✓ Static balance: disc rests in random position when mounted on arbor
— no preferred heavy side

VCE-T05 — Stepped Speed Pulley × 2

ROUGH DRAFT

Lathe Instructions — Brass CZ121 or Aluminium 6061

DO NOT MACHINE UNTIL SPRING MOTOR IS DELIVERED

Step diameters cannot be calculated until the spring motor output shaft RPM is confirmed under operating load. Machine the stepped pulley only after the remontoire spring motor (VCE-T02) has been delivered and its output RPM measured. See VCE-CPR-001 for the calculation method.

Once motor output RPM is known, calculate the three step diameters for 33⅓, 45, and 78 RPM using: $\text{step diameter} = (\text{target RPM} \times \text{platter drive rim circumference}) \div (\text{motor shaft RPM} \times \pi)$. Verify the calculation before cutting.

Machining Sequence

- 1 Source brass CZ121 or aluminium 6061 round bar, diameter sufficient for the largest step plus 5mm, length sufficient for all three steps plus facing allowance.
- 2 **Face and centre both ends.** Support with tailstock centre for all turning operations.
- 3 **Turn each step to calculated diameter** , largest step first (nearest chuck), smallest last. Leave 0.2mm finishing allowance on each step.

- | | | |
|---|---------------------------------------|---|
| 4 | Finish each step. | 0.2mm finishing pass. Steps must be concentric to within 0.02mm — re-chuck between steps is not acceptable. Complete all steps from one chucking. |
| 5 | Machine belt groove profile | on each step. Groove width and depth match the belt specification. Use a form tool ground to the belt profile, or a parting tool with a small radius at root. |
| 6 | Bore and ream motor shaft bore | to match motor shaft diameter. H7 tolerance. Keyway if required by motor shaft — broach after boring. |
| 7 | Deburr all groove edges. | A burr in a belt groove will damage the belt within minutes of running. |

ROUGH DRAFT –
CONFIRM BELT
PROFILE BEFORE
CUTTING
GROOVES

ACCEPTANCE CHECKS

- ✓ All three step diameters within $\pm 0.1\text{mm}$ of calculated values
- ✓ All steps concentric to bore within 0.02mm TIR
- ✓ Belt grooves free of burrs — belt seats fully in each groove without riding the edges
- ✓ Verify RPM on each step with strobe disc after installation — confirm 33⅓, 45, 78 RPM within $\pm 0.5\%$

VCE-T06 — Mu-Metal Bearing Containment Ring × 2

ROUGH DRAFT

Sheet Metal Shear + Roll + Lathe Finish

ANNEAL AFTER FORMING – THIS IS NOT OPTIONAL

Mu-metal loses the majority of its magnetic shielding effectiveness when cold-worked. Bending, rolling, or cutting mu-metal sheet destroys the magnetic domain alignment that gives it shielding properties. The ring must be annealed in a hydrogen atmosphere at 1,100°C after forming and before installation. Without annealing, the ring will not achieve the required $<5\mu\text{T}$ field at cartridge position. Arrange annealing with a specialist heat treatment contractor or source a mu-metal fabricator who includes post-forming annealing in their service.

Fabrication Sequence

- 1 Source mu-metal sheet 1.5mm thick from a specialist supplier (Magnetic Shield Corp, Amumetal, or equivalent). Request certification of permeability specification.
- 2 **Cut strip** to length = $\pi \times (\text{bearing assembly OD} + 4\text{mm clearance}) \times 1.05$ (overlap allowance). Width = bearing assembly height + 20mm (10mm overlap each end).
- 3 **Roll to cylinder** on a sheet metal roller. Do not use brake bending — the ring must be a true cylinder, not a polygon. Target: inner diameter = bearing assembly OD + 2mm radial clearance each side.
- 4 **Tack weld seam** with stainless TIG weld — minimum heat input. Do not use carbon steel filler. Three short tack welds, evenly spaced.
- 5 **Send for annealing** — hydrogen atmosphere, 1,100°C, slow cool. Do not proceed to installation until annealed.
- 6 **After annealing: light lathe skim of inner and outer diameter** if roundness is poor. Minimum material removal — do not cold-work the annealed surface more than necessary. 0.1mm skim only.
- 7 **Verify field attenuation with gaussmeter** before installation. Assemble bearing with ring in place. Measure field at cartridge position. Must be $<5\mu\text{T}$. If not, add a second layer of mu-metal sheet and re-anneal.

ACCEPTANCE CHECKS

- ✓ Ring is a true cylinder — roundness within 1mm over full circumference
- ✓ Ring has been annealed — confirmed by heat treatment certificate
- ✓ Field at cartridge position with ring installed: $<5\mu\text{T}$ (measured with calibrated gaussmeter)
- ✓ Ring fits around bearing assembly with 1.5–2.5mm radial clearance — does not contact bearing

not need the precision tolerances of the turntable platter, but must be dynamically balanced — an unbalanced flywheel at generator speed creates vibration that may couple to the turntable through the console structure.

Machining Sequence

- 1

Source cast iron disc (preferred) or mild steel plate, 210mm minimum diameter, 35mm minimum thickness. Cast iron preferred for its natural vibration damping properties.
- 2

Face both sides

to establish parallel reference faces. 0.5mm finishing pass.
- 3

Turn outer diameter to 200mm.

Concentric with bore datum. 1.6µm Ra finish on outer diameter.
- 4

Bore generator shaft hole

to match generator shaft OD — confirm generator shaft diameter before machining. H7 tolerance. Keyway if required.
- 5

Dynamic balance.

Mount on arbor, run on dynamic balancing machine. Target: <2g·cm. Remove material by drilling shallow holes in the rim face — do not reduce rim diameter.

Machining Parameters

ROUGH DRAFT

OPERATION	PARAMETER	NOTES
TURNING – CAST IRON	400 RPM · 0.2mm/rev · carbide insert · dry	Cast iron machines dry. Flood coolant causes thermal cracking in some grades.
FACING – CAST IRON	500 RPM · 0.15mm/rev · 0.5mm DOC · dry	Watch for hard spots — reduce DOC if surface is irregular.
BORING – CAST IRON	600 RPM · 0.1mm/rev · boring bar · dry	Single boring pass before reaming for best roundness.

ACCEPTANCE CHECKS

- ✓ Bore diameter and tolerance match generator shaft specification
- ✓ Dynamic balance: <2g·cm on balancing machine
- ✓ No surface porosity visible on machined faces — cast iron with significant porosity is unusable
- ✓ Flywheel spins freely on generator shaft with no detectable radial play

Lathe Instructions — Brass CZ121

Sixteen small brass parts in two designs. Batch production — machine all sixteen from the same bar stock in one session to ensure consistency. The brass cups must be identical in depth and diameter to ensure the platform sits level. The spring housings must have matched thread to allow height adjustment for levelling.

Brass Cups (VCE-P01) — Machining Sequence

- 1 Source brass CZ121 round bar, 30mm diameter. Cut 8 billets, 20mm length each.
- 2 **Face and turn** to 28mm. All eight pieces to same diameter within
outer diameter $\pm 0.05\text{mm}$ — measure each with micrometer.
- 3 **Bore** 18mm diameter, 9mm deep. Ball seats half-proud. Recess
ball must be concentric to outer diameter — use boring bar from
recess. same chucking as outer diameter turning.
- 4 **Drill and tap** in base — $M6 \times 1.0$, through to recess bottom.
mounting holes Two holes per cup, 180° apart.
- 5 **Polish** 600-grit abrasive on a 17mm wooden dowel. Ball bearing
recess. must move freely in cup — test with actual ball bearing
before finalising all eight cups.

Spring Housings (VCE-P02) — Machining Sequence

- 1 Source brass CZ121 round bar, 30mm diameter. Cut 8 billets, 40mm length each.
- 2 **Turn outer** to match spring OD plus 2mm wall. Spring OD
diameter determined after spring procurement.
- 3 **Thread** $M20 \times 1.5$ for height adjustment. All eight to identical
outer thread specification — check with thread gauge.
diameter
- 4 **Bore** to spring OD + 0.5mm clearance. Depth: spring free length
spring minus 5mm. Spring must seat squarely — face of bore must
seat be perpendicular to bore axis within 0.1mm.
- 5 **Machine** on outer diameter for spanner — 27mm A/F hexagon on
flats lower section for adjustment.

SPRING RATE MUST BE CALCULATED BEFORE ORDERING SPRINGS

Do not order springs or machine spring housings until the total mass resting on the platform has been measured. Weigh the assembled plinth, 50mm slate slab, and turntable as a unit. Calculate required spring rate for 1.5Hz resonance: $k = m(2\pi f)^2$ where m is mass per spring (total mass ÷ 4) and f = 1.5Hz. Source springs to that rate ±10%.

ACCEPTANCE CHECKS – BOTH PARTS

- ✓ All 8 brass cups: recess depth identical within ±0.1mm (measure with depth gauge)
- ✓ Steel ball bearing seats and rolls freely in each cup
- ✓ All 8 spring housings: thread engagement smooth on M20 × 1.5 nut — no binding
- ✓ Spring seats square — spring sits perpendicular to housing axis when assembled
- ✓ Platform assembly: resonant frequency 1.4–1.6Hz when loaded (test by gentle excitation)

VCE-OD02 — Audience Unit Flywheel × 1 (Option D only)

ROUGH DRAFT

Lathe Instructions — Cast Iron

Identical to VCE-G01 in fabrication method, but smaller. The audience unit flywheel stores less energy than the console flywheel — its purpose is simply to smooth hand crank strokes into consistent generator rotation, not to bridge the gap between strokes as critically as in the console.

PARAMETER	VALUE	NOTES
MATERIAL	Cast iron	<i>Same as VCE-G01</i>
DIAMETER	100mm	<i>Half the console flywheel — no turntable to protect</i>
MASS TARGET	0.5kg	<i>Estimated — verify on first article</i>
BALANCE TOLERANCE	<2g·cm dynamic	<i>Same as console flywheel</i>
BORE	Match generator shaft OD	<i>Same generator spec as</i>

— confirm before
machining

console

Machining parameters and acceptance checks: identical to VCE-G01. Refer to VCE-G01 instructions above.

SECTION III

Casting (*Option C Only*)

VCE-OC01 — Gravity Weight × 1 (Option C only)

ROUGH DRAFT

Grey Iron Casting + Lathe Finish

SOURCE FIRST — CAST ONLY IF NECESSARY

A 40kg cast iron cylinder at 200mm diameter is a standard industrial counterweight size. Check with industrial counterweight suppliers before committing to a casting. Sourcing a machined counterweight is significantly cheaper and faster than commissioning a casting. If sourcing: verify material is grey iron (not steel), verify mass is 40kg ±1kg, and verify the cylinder diameter allows guide rail pad machining at 200mm OD with adequate wall thickness.

If Casting Is Required

- 1 Prepare a simple cylindrical pattern — 202mm diameter (2mm machining allowance), length to achieve 40kg in grey iron (density ~7,200 kg/m³ — calculate required length).
- 2 Commission a sand casting from a foundry. Grey iron grade 250 or equivalent. Specify: no significant porosity on outer surfaces, no cold shuts, no shrinkage cavities visible on first 5mm from surface.
- 3 Inspect casting before machining. Reject if any surface defect penetrates more than 3mm — the guide rail pad area must be sound metal.
- 4 **Lathe — face both ends** square to cylinder axis. Perpendicularity: <0.2mm per 200mm.

- | | | |
|---|---|--|
| 5 | Lathe — turn outer diameter to 200mm. | Concentricity: outer diameter to bore within 0.5mm — guide rail pads are machined from this surface. |
| 6 | Mill guide rail pads. | Two brass pads (or machined iron flats) on opposing sides of the cylinder, sized to run in the guide rail channels with <0.5mm lateral clearance. This is the safety-critical dimension. Verify fit in actual guide rails before installation. |
| 7 | Drill and install cable attachment insert. | Top face, centre. M16 stainless steel threaded insert, Loctite 638 retaining compound. Pull-out strength test: apply 400kg tensile load to insert before final installation. Do not skip this test. |

CABLE ATTACHMENT IS SAFETY CRITICAL

This insert carries 40kg continuously, with dynamic shock loads during lift cycles. A pull-out failure drops 40kg from up to 2 metres. Test the insert to 400kg tensile load (10:1 safety factor on 40kg). Inspect visually after test — any movement of the insert during testing is a rejection. Use a new insert.

ACCEPTANCE CHECKS

- ✓ Mass: 40kg ±1kg (weigh on calibrated scale)
- ✓ Guide rail pads: <0.5mm lateral play in actual guide rails
- ✓ Weight descends vertically in guide rails with no binding or jamming over full 2m travel
- ✓ Cable attachment insert: 400kg tensile test passed
- ✓ No casting defects visible on machined surfaces

SECTION IV

Steel Fabrication *(Option C Only)*

VCE-OC04 — Guide Rail Tower Structure × 1 (Option C only) ROUGH DRAFT

Structural Steel Fabrication + Walnut Veneer Cladding

The guide rail tower is the tallest structural element in the instrument at 2,600mm. It must be rigid enough that bass energy from the corner horn,

transmitted through the floor, does not cause vibration of the tower structure that could couple back to the generator through the cable. The primary structural frame is mild steel hollow section. Walnut veneer cladding is applied after structural assembly.

Structural Frame

- 1

 Source mild steel rectangular hollow section (RHS), 50mm × 50mm × 3mm wall. Cut four uprights to 2,500mm. Cut horizontal cross-members for top, bottom, and two intermediate positions.
- 2

Fabricate rectangular frame — two uprights joined by cross-members at top and bottom, with intermediate cross-members at 800mm and 1,600mm height. TIG or MIG weld. Full penetration welds on all structural joints.
- 3

Machine guide rail channels into uprights at correct spacing to match weight guide pads — or weld precision guide rail brackets to uprights and fit brass guide rails separately. Second method preferred — allows guide rail replacement without structural work.
- 4

Weld floor plate. 300mm × 300mm × 10mm mild steel base plate with four 12mm anchor bolt holes. Floor anchor bolts are mandatory for safety.
- 5

Weld apex pulley bracket. Heavy wall bracket at tower top, centred. The pulley carries the full 40kg cable load — weld with full penetration and gusset plates.
- 6

Grind all welds smooth on visible faces. Structural welds on non-visible faces may be left as-welded.
- 7

Apply primer and undercoat before cladding — the steel will not be accessible after veneer is applied.

Walnut Veneer Cladding

- 1

 Commission 18mm Baltic birch ply panels cut to fit each face of the tower — commission with the cabinetmaker for the turntable plinths (VCE-C01) to ensure finish consistency.
- 2

 Bond panels to steel frame with construction adhesive and screws. Countersink all screws — fill with walnut filler before finishing.
- 3

 Apply walnut veneer and satin lacquer finish — matching VCE-C01, VCE-C02, VCE-C03.

4

Glass inspection panel on front face

— borosilicate glass, brass frame, 150mm × 1,800mm window to show weight descent. Fit after structural assembly and before veneer on that face.

STRUCTURAL INTEGRITY CHECK BEFORE USE

Before installing the weight, apply a static test load of 80kg (2× working load) to the cable attachment point via the pulley, with the tower anchored to the floor. Hold for 10 minutes. Inspect all welds and anchor points. Any visible deflection, cracking, or movement is a rejection — do not install the weight until the failure is found and repaired.

ACCEPTANCE CHECKS

- ✓ Tower is vertical within 2mm over full height (check with level)
- ✓ Floor anchor bolts installed — tower does not tip under 50kg lateral force at apex
- ✓ Weight descends and lifts smoothly over full 2m travel with no binding
- ✓ 80kg static load test passed on cable attachment
- ✓ No structural movement when bass horn is playing at full volume — check by touch during test

SECTION V**Electronics *Self-Build*****QUALIFIED TECHNICIAN – ALL OF THIS SECTION**

Every assembly in this section involves voltages that are lethal at available current levels. 600V DC from the capacitor bank, 250V DC for field coil supply, 150–200V DC for valve anodes. All work in this section must be performed by or under the direct supervision of a qualified high-voltage electronics technician. All high-voltage circuits must be fully enclosed with panel interlocks. The bleed resistor must be installed before any other connection to the capacitor bank is made.

Electronics Assembly Instructions

The Cockcroft-Walton multiplier is a passive ladder of diodes and capacitors converting 24V AC to 600V DC. It contains no active components — no transistors, no ICs, no switching elements. Failure modes are all passive: diode or capacitor breakdown under overvoltage, or solder joint failure. The circuit is well-understood and reliable if built with correctly-rated components and good solder joints.

Component Specification

COMPONENT	SPECIFICATION	NOTES
DIODES	1N4007 or equivalent — 1kV PIV, 1A	<i>1kV minimum PIV. Low reverse leakage. Standard rectifier diode.</i>
CAPACITORS — MULTIPLIER	100µF, 1kV DC — polypropylene or film	<i>Must be 1kV rated. Do not use electrolytic in the ladder — polarity reversal during multiplier action.</i>
BLEED RESISTOR	1MΩ, 5W wirewound — permanently across output	<i>Mandatory safety component. Discharges bank in 60 seconds if disconnected.</i>
ZENER CLAMP (OPTION D)	620V zener, 5W — across output rail	<i>Mandatory if Option D installed. Prevents overvoltage from dual charging sources.</i>
BLOCKING DIODE (OPTION D)	1kV, 1A — in series with each C-W output before bank junction	<i>Prevents reverse current between C-W units.</i>
PCB SUBSTRATE	FR4, 2mm, 2kV dielectric rating minimum	<i>Standard PCB material is adequate at 600V with correct creepage distances.</i>
CREEPAGE DISTANCE	6mm minimum between conductors at 600V	<i>IEC 60664-1 requirement for 600V. Do not reduce this spacing.</i>

Assembly Sequence

- 1 Design PCB layout** with 6mm minimum creepage between all conductors at 600V potential difference. Have layout reviewed by the qualified technician before fabricating PCB.
- 2 Install diodes.** Observe polarity carefully — one incorrectly-oriented diode will prevent the circuit from functioning and may damage other diodes. Check each diode orientation with a continuity tester before soldering the next stage.
- 3 Install capacitors.** Non-polarised film capacitors — orientation is not critical but lead dress is. Keep leads short and component bodies away from the PCB surface for heat dissipation.
- 4 Install bleed resistor FIRST** before any other output-side connections. This is the mandatory safety element — it must be in place before the circuit is energised for the first time.
- 5 Install in mu-metal shielded housing.** All connections to and from the housing via strain-relieved high-voltage rated cable (1kV minimum insulation rating).
- 6 Test with dummy resistive load before connecting to capacitor bank.** Apply 24V AC from a bench supply. Measure output voltage. At no-load, output should be approximately $24V \times (\text{number of stages} \times 1.4)$. Verify this matches the calculated target before connecting the capacitor bank.
- 7 Connect bleed resistor across capacitor bank terminals** before connecting the C-W output to the bank.

COMMISSIONING CHECKS

- ✓ Output voltage with dummy load: within 10% of calculated 600V target
- ✓ Bleed resistor installed and verified — output discharges to <50V within 60 seconds of disconnecting input
- ✓ No component runs hot during 10-minute test with dummy load — any warm component is suspect
- ✓ Output ripple at cranking speed: measure with oscilloscope — excessive ripple indicates a failed capacitor stage
- ✓ If Option D: both C-W units charge bank simultaneously without instability — bank voltage stable, neither unit backfeeds the other

High-Voltage Electronics Assembly Instructions

The capacitor bank is the instrument's primary energy reservoir — its modern Leyden jar. It stores 4,000–5,000 joules at 600V DC. The bank is assembled from individual high-voltage ceramic capacitors in series-parallel combination to achieve the required total capacitance at 600V rating.

Capacitor Selection

PARAMETER	REQUIREMENT	NOTES
VOLTAGE RATING	1kV DC minimum per cell	20% safety margin above 600V operating voltage
CAPACITANCE PER CELL	Determine from series-parallel arrangement to achieve 25,000µF total at 600V	Calculated: $2E/V^2 = 2 \times 4500 / 360000 = 25mF$ total
DIELECTRIC	Barium titanate ceramic (X7R or better) or polypropylene film	Self-discharge measured in years. Do not use electrolytic.
ESR	Low ESR specified – <100mΩ per cell at 1kHz	High ESR generates heat during charging. Measure before installation.

Assembly Notes

- 1

Balance series-connected capacitors

with equalising resistors across each capacitor (100kΩ, 1W) — ensures equal voltage sharing and prevents one cell from seeing more than its rated voltage.
- 2

Install bank in mu-metal shielded housing

with interlocked access panel — panel must disconnect bleed path and add shorting bar across bank terminals when opened.
- 3

Measure actual stored energy

by charge-discharge test — charge bank to 600V, disconnect source, discharge into known resistive load, integrate $V \times I \times dt$. Actual stored energy should be 3,800–5,200J.
- ESTIMATED RANGE

- 4

Label all external surfaces with voltage warning — high-voltage warning labels in multiple languages, minimum 50mm × 50mm, on all faces adjacent to any access point.

Two independent signal channels, each: SUT input → V1 (12AX7) → passive RIAA network → V2 (12AX7) → output to TVC. Powered from the capacitor bank through dropping resistors. This is a conventional single-ended Class A valve phono stage operating at high voltage from an unusual supply. Any experienced valve technician will recognise the circuit topology.

Component Specification — Critical Items

COMPONENT	SPECIFICATION	NOTES
VALVES	12AX7 (ECC83) — low-noise selected	<i>Cherry-pick lowest noise specimens from a batch. Test all candidates and select the four quietest for V1 positions (first stage — most noise-sensitive).</i>
RIAA RESISTORS	0.1% tolerance metal film — IEC 60098 time constants	<i>Standard RIAA: 3180µs, 318µs, 75µs. Do not substitute 1% resistors in the RIAA network.</i>
RIAA CAPACITORS	Polypropylene film — 1% or better tolerance	<i>Polypropylene for low distortion and long-term stability.</i>
HEATER SUPPLY	6.3V DC regulated — not AC	<i>DC heater eliminates 6.3V hum from heater-to-cathode leakage. Mandatory for low-noise operation.</i>
ANODE RESISTORS	Metal film, 1% tolerance, 2W minimum rating	<i>2W rating for thermal headroom — actual dissipation well below this.</i>
BYPASS CAPACITORS	Polypropylene film on all cathode bypass	<i>Electrolytic acceptable here but polypropylene preferred for long-term reliability.</i>

Assembly and Commissioning Sequence

- 1 **Build chassis.** Aluminium or steel chassis, all valve retainers installed, point-to-point wiring or PCB — builder's choice. Point-to-point preferred for serviceability and repairability in the field.
- 2 **Install dropping resistors for anode supply** from 600V rail to valve anode voltage (target: 150–200V at valve anode — calculate dropping resistor from supply voltage and target current).
- 3 **Wire RIAA networks.** Use shielded wire for all signal paths. Keep input wiring (cartridge level) short and away from power wiring.
- 4 **First power-up: heater supply only.** Verify 6.3V DC $\pm 5\%$ at all valve heater pins before applying anode voltage. Incorrect heater voltage will damage valves.
- 5 **Apply anode voltage.** Measure anode voltage on all valves. Target: 150–200V depending on dropping resistor calculation. Measure quiescent current: 1.0–1.5mA per stage. Adjust cathode resistor if current is outside range.
- 6 **Measure RIAA frequency response** with audio signal generator. Inject signal at SUT input, measure output. RIAA curve must be within $\pm 0.5\text{dB}$ from 20Hz to 20kHz. If outside spec: check RIAA component values with precision meter before assuming circuit fault.
- 7 **Measure noise floor.** Short SUT primary input. Measure output noise A-weighted. Target: below vinyl surface noise level — approximately $< 0.1\mu\text{V}$ equivalent input noise. If noisier: try different 12AX7 specimens in V1 position first (valve selection is the primary noise variable).
- 8 **Measure gain.** Total gain at 1kHz after RIAA should be approximately 52dB. Verify both channels match within $\pm 1\text{dB}$.

COMMISSIONING PASS CRITERIA

- ✓ Heater voltage: 6.3V DC $\pm 5\%$ on all valves
- ✓ Anode voltage: 150–200V on all valves (within $\pm 10\%$ of design target)
- ✓ Quiescent current: 1.0–1.5mA per stage
- ✓ RIAA response: $\pm 0.5\text{dB}$ from 20Hz–20kHz on both channels
- ✓ Total gain at 1kHz: approximately 52dB, channels matched within

±1dB

- ✓ Noise floor: below subjective vinyl surface noise level on playback
- ✓ Channel separation: >60dB at 1kHz
- ✓ No oscillation — check with oscilloscope at output with no input signal

VCE-E07 — PLL Speed Control Circuit × 2

ROUGH DRAFT

Low-Voltage Electronics Assembly

The PLL circuit compares actual platter speed against a 32.768kHz quartz crystal reference and adjusts the eddy current braking load to maintain speed within 0.05% WRMS. It operates at 12V from the capacitor bank via voltage regulation — the only low-voltage circuit directly powered from the high-voltage rail. Power consumption: 30mW. This is a small, well-understood circuit.

COMPONENT	SPECIFICATION	NOTES
CRYSTAL REFERENCE	32.768kHz quartz — watch crystal grade	<i>Post-1921 technology. If strict Victorian authenticity preferred, substitute tuning fork oscillator — W&F degrades to <0.15% WRMS.</i>
SPEED SENSOR	Hall effect or optical — triggered by reference marks on platter underside	<i>8 reference marks for 33⅓ RPM gives adequate phase resolution.</i>
PHASE COMPARATOR	Standard CMOS PLL IC — CD4046 or equivalent	<i>Well-documented, easily sourced. Operates at 5V or 12V.</i>
BRAKING CONTROL	Variable resistor across eddy current governor — controlled by PLL output	<i>Increasing resistance reduces braking — platter speeds up. Decreasing resistance increases braking — platter slows down.</i>
SUPPLY ISOLATION	DC-DC isolated converter — 600V rail to 12V output	<i>Isolates the low-voltage PLL circuit from the high-voltage rail. Mandatory for safety.</i>

COMMISSIONING CHECKS

- ✓ PLL locks to crystal reference within 5 seconds of platter reaching approximate target speed
- ✓ Wow and flutter: <0.05% WRMS measured with test disc and flutter meter
- ✓ Speed stable across full main spring wind range — does not drift as spring depletes
- ✓ Speed correct on all three pulley steps: 33⅓, 45, and 78 RPM (verify with strobe disc)
- ✓ No audible 30mW supply noise in signal chain — measure noise floor with PLL powered vs unpowered

VCE-E08 — Peripheral Light Indicator Circuit × 1 (Option B+ only) ROUGH DRAFT

Low-Voltage Electronics Assembly

Three jewelled indicator lenses — green, amber, red — showing capacitor bank charge state. Powered from a 1,000µF buffer capacitor trickle-charged from the main bank. The buffer ensures the red warning remains lit for 30–60 seconds after the main bank is exhausted — precisely when the warning is most needed.

COMPONENT	SPECIFICATION	NOTES
INDICATORS	5mm high-brightness diffused LED × 3 (green, amber, red)	<i>Diffused lens for wide viewing angle — operator views from oblique angle.</i>
LENS BEZELS	12mm jewelled brass bezels with diffused glass — machine apertures in console panel	<i>Victorian signal lamp aesthetic. Brass bezels machined to console panel.</i>
THRESHOLD — GREEN	Bank above 75% charge — zener reference to bank voltage divider	<i>Calibrate zener threshold during commissioning against actual bank voltage.</i>
THRESHOLD — AMBER	Bank 40–75% charge	<i>Overlap is not acceptable — thresholds must be mutually exclusive.</i>
THRESHOLD — RED	Bank below 40% charge	<i>At 40% charge, approximately 90</i>

		<i>seconds of operation remain at 20W draw.</i>
BUFFER CAPACITOR	1,000 μ F at appropriate voltage — trickle charged from main bank via 10k Ω resistor	<i>Holds red LED on for 30–60 seconds after main bank exhaustion.</i>

COMMISSIONING CHECKS

- ✓ Green illuminates when bank is charged above 75% of nominal — verify against voltmeter reading
- ✓ Amber illuminates at 40–75% — no overlap with green or red
- ✓ Red illuminates below 40% — and remains lit for minimum 30 seconds after main bank disconnected
- ✓ Indicators visible from operator position in normal ambient and low-light conditions
- ✓ No false triggering — indicator does not flicker during normal operation

WHEN YOU HAVE BUILT THIS INSTRUMENT

The Volta Confluence Engine has never been physically built. You will be the first. Your results — which specifications held, which needed adjustment, which parameters proved correct, and which were wrong — are valuable to every builder who follows. Please complete and submit a Builder's Report (document VCE-BRT-001). Your findings will update the confidence levels in this document and improve the experience of every subsequent builder. The certainty labels throughout this document will only move from Rough Draft to Confirmed through the work of people like you.

INSTRUMENT

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Powered Two-
Channel
Phonograph
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DOCUMENT

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